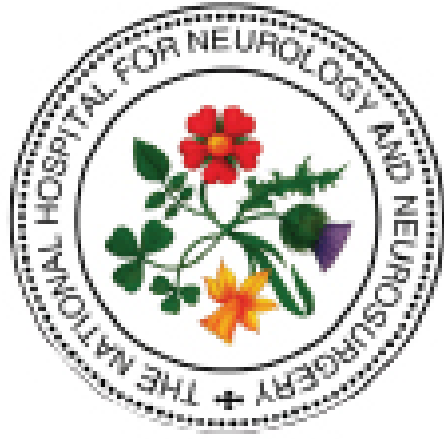




► Brain Tissue Oxygen ($P_{bt}O_2$) Guided Management of Treated Aneurysmal Subarachnoid Haemorrhage (aSAH)

Monitoring of brain tissue oxygenation is an important part of multimodality monitoring (MMM). Continuous monitoring of $P_{bt}O_2$ is an additional tool to enable detection of impaired cerebral blood flow after SAH – it allows the early identification of patients who are at increased risk of delayed cerebral ischaemia (DCI), using medical management by manipulating SBP and O_2 and timely radiological intervention to help prevent infarction. It can help identify the need for and the effect of various therapies such as hyperoxia, induced hypertension and targeted temperature management in SAH patients.

ICP >25mmHg on probe insertion

- CT scan required to exclude mass effect or hydrocephalus
- Subsequent **sustained ICP >22mmHg** should prompt further imaging
- ICP trend** more relevant than individual readings in 'awake' patients (especially if agitated) in combination with GCS

2 hours after probe insertion

- Increase FiO_2 to 100% for 5min – *this is of limited value in awake patients*
- Check for increased $P_{bt}O_2$ within 1-2min (*may take longer in some cases*)
- If $P_{bt}O_2$ increases within 1-2min, catheter likely to be in position
- If no change in $P_{bt}O_2$ within 1-2min consider CT scan to confirm position
- If no $P_{bt}O_2$ response to 5min hyperoxia challenge, and CT scan shows adequate position, catheter may be faulty – consider new catheter

Neurogenic temperature management

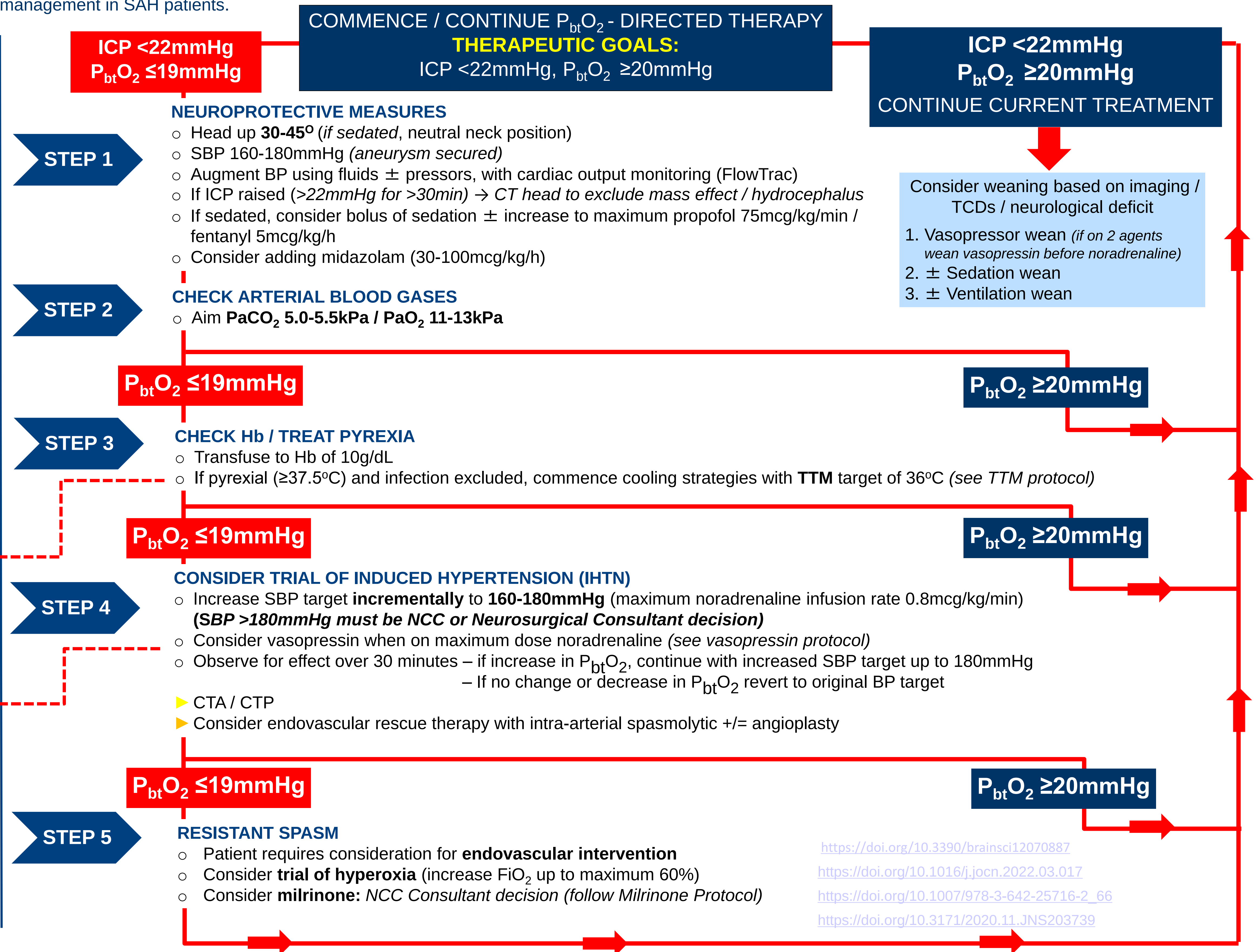
- Pharmacological:** Paracetamol
- Non-pharmacological:** Targeted Temperature Management (TTM) (see protocol)

Induced hypertension (IHT)

- BP augmentation known to be effective therefore should continue even when $P_{bt}O_2$ adequate
- Noradrenaline >1mcg/kg not effective in SAH vasospasm
- Consider hydrocortisone 50mg tds if vasopressor requirement high

Class of Recommendation

- **Class 1: STRONG**
Recommended (Benefit >>> Risk)
- **Class 2a: MODERATE**
Reasonable (Benefit >> Risk)
- **Class 2b: WEAK**
May be reasonable (Benefit ≥ Risk)
- **Class 3: STRONG**
Potential harm (Risk > Benefit)



<https://doi.org/10.3390/brainsci12070887>

<https://doi.org/10.1016/j.jocn.2022.03.017>

https://doi.org/10.1007/978-3-642-25716-2_66

<https://doi.org/10.3171/2020.11.JNS203739>